



Short communication

Risk mitigation for indoor air quality using the example of construction products – efforts towards a harmonization of the health-related evaluation in the EU

Ana Maria Scutaru^{a,*}, Thomas Witterseh^b

^a German Environment Agency (UBA), Berlin, Germany

^b Danish Indoor Climate Labelling, Danish Technological Institute, Taastrup, Denmark

ABSTRACT

In Europe, the Construction Products Regulation sets harmonized conditions for the marketing of construction products with the objective of protecting the building users' health. Until now only three European countries have implemented requirements for the assessment of VOC emissions from construction products. Therefore, the European Commission is planning the issue of a delegated act on the communication of VOC emissions from construction products in the form of VOC classes. A key prerequisite for defining the VOC classes is the completion of the EU-LCI list currently being carried out by a group of experts from ten European countries. This paper reports on the development of the VOC class concept, the progress of the EU-LCI harmonization framework and Germany's current efforts to ensure a high level of health protection for building users and avoid dangers from construction product emissions.

1. Introduction

Construction products, including items such as floorings, particle boards, wooden panels, carpets and adhesives, are known to emit a variety of gaseous substances described as volatile organic compounds (VOC). Measurements in indoor environments have shown that many VOC occur in concentrations considerably higher than in ambient air (Wolkoff, 1995; Paciência et al., 2016). Several organic compounds have been considered to exert adverse effects upon human health at the levels found in indoor air (e.g., Beall and Ulsamer, 1981; Mølhave, 1991).

The use of low emission construction products is a key measure to facilitate good indoor air quality. This has become an even more prominent issue in the context of modern construction, particularly in energy-efficient buildings where the rate of air exchange with fresh ambient air may be limited (Seppänen, 2008; Müller and Mertes, 2017).

In the European Union (EU), the Construction Products Regulation (CPR) was set out to harmonize test methods and product labeling, aiming to remove trade barriers for all construction products on the European market (Construction Products Regulation, 2011). The CPR defines seven basic requirements for construction works and the building products incorporated therein. Requirement no. 3 "Hygiene, Health and Environment" stipulates that emissions of dangerous substances, volatile organic compounds from construction products must not endanger the health of building users.

Under CPR manufacturers must apply the CE marking to their products covered by a harmonized European standard (hEN). hEN defines the parameters to be described in a so-called product performance declaration. In the past the CE marking, a premise for many products to access the European market, had been used to characterize product properties such as mechanical stability and flammability. Within the CPR it is planned to supplement the CE marking by information regarding the emission of so-called dangerous substances. Currently, only very few product standards in the EU include health requirements. Emission of VOC, in particular, have not been addressed with the exception of formaldehyde. In the future, the CE marking will be accompanied by a performance declaration of the product regarding VOC emissions using classes. For this purpose, the European Member States will have to define their national requirements - the accepted level of emissions - for allowing products on their markets.

Germany has been the first EU country to provide a mandatory scheme for the health-related evaluation of VOC emissions from construction products, since 2006. This so-called AgBB scheme was conceived by the Committee for health-related evaluation of building products (*Ausschuss zur gesundheitlichen Bewertung von Bauprodukten; AgBB*) (AgBB, 2018; Däumling, 2012). France and Belgium established corresponding regulations in 2011 and 2014, respectively (Decree n° 2011-321, 2011; Decree [C - 2014/24239], 2014). These three mandatory systems use, however, different lists of individual substances for the

* Corresponding author.

E-mail address: anamaria.scutaru@uba.de (A.M. Scutaru).

<https://doi.org/10.1016/j.ijheh.2020.113588>

Received 8 February 2020; Received in revised form 27 April 2020; Accepted 28 May 2020

Available online 9 August 2020

1438-4639/© 2020 Published by Elsevier GmbH. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

assessment of the VOC emissions from construction products: Germany's AgBB committee developed pass-fail criteria for a comprehensive list of 185 substances. France requires the declaration of an emission class based on a selection of 10 indoor pollutants. In Belgium construction products have to comply with a set of criteria similar to the German system, specifically referring to the harmonized long list of EU-LCI values (LCI = 'lowest concentrations of interest') described below.

The European Commission has recognized the insufficient implementation of health requirements in the CPR. It therefore plans to issue a delegated legal act on how to communicate construction product emissions in terms of VOC classes. As early as 2005, the European Commission issued a mandate to the European Committee for Standardization (CEN) for the development of horizontal standardized methods for the testing of construction products in environmental chambers, which lead to the establishment of the CEN Technical Committee CEN/TC 351 'Dangerous Substances'. The objective of test chamber measurements is to characterize the types and quantities of dangerous substances released under standardized conditions.

Besides the mandatory systems there are a number of voluntary, health-related quality labels in European countries offering consumers assistance in choosing a 'safe' product regarding controlled emissions. Their characteristics have been described and compared first in 2005 (ECA Report 24, 2005). These voluntary labeling systems have evolved over the years taking into consideration accumulated experience and market development.

One example is the voluntary Danish Indoor Climate Label scheme, introduced in 1993 in Denmark to encourage development of low-emitting building materials and products used in the indoor environment. Although it was originally intended for the Danish market, manufacturers from several European countries are now holding labeling licenses, and the label is used and recognized world-wide. During the first 10 years of operation of the labeling scheme the need for an internationally harmonized assessment method for material emissions had already been identified.

2. Progress of the harmonization activities and current challenges

In 2007 the four major European labels joined an initiative to harmonize the assessment criteria for product evaluation. Their recommendations for common core and transitional criteria for a European labeling scheme are published in the ECA Report 27 (ECA Report 27, 2012).

The core criteria specify parameters upon which the expert group reached a consensus. These are:

- (1) Assessment of all VOC emitted by a product using harmonized toxicological reference values (EU-LCI values): In practice, the test chamber concentration C_i of each individual compound is compared against its corresponding EU-LCI value LCI_i , yielding a substance-specific index R_i :

$$R_i = C_i / LCI_i \quad (1)$$

The concept of R_i was first introduced by the ECA Report 18 (1997a). The smaller R_i , the lower the risk for potential health effects emerging from emissions of that substance. For $R_i < 1$ and exposure to that single substance, no adverse health effects are expected. However, construction products emit mixtures of multiple compounds. The body of toxicological evidence suggests the usefulness of a linear additive model. Consequently, a cumulative hazard index R was devised, which is calculated as follows:

$$R = \text{sum of all } R_i = \text{sum of all ratios } (C_i / LCI_i) \leq 1 \quad (2)$$

To avoid adverse health effects, this R value shall not exceed the value 1.

- (2) Restriction of carcinogens classified as 1A and 1B (according to Regulation (EC) No 1272/2008, 2008)
- (3) Restriction of the sum of VOC (TVOC: total volatile organic compounds).

Additional parameters such as restriction of semi-volatile organic compounds (SVOC) and substances without EU-LCI values and sensory testing were considered in 2012 as transitional criteria for which consensus is still to be reached. The publication of the ECA Report 27 marked the end of phase I of the harmonization initiative of construction products emissions.

During phase II two workshops were organized by the Joint Research Centre (JRC) of the European Commission. The first workshop in June 2010 was supposed to foster further discussion on the core and transitional criteria for a European labeling scheme in an enlarged group of representatives of other voluntary labeling schemes. But this part of the harmonization process has not been pursued any further yet. The second workshop organized in September 2010 was the kick-off meeting for the development of the EU-LCI values.

2.1. Establishment of EU-LCI values

In order to provide a common basis for the assessment of building product emissions, experts from ten European countries started developing the EU-LCI list in 2011 under the coordination of JRC. The experts defined EU-LCI values as 'health-based reference concentrations for inhalation exposure used to assess emissions after 28 days from a single product during a laboratory test chamber procedure' (ECA Report 29, 2013). The process of establishing EU-LCI values started with the compilation of a list of substances emitted by construction products in detectable quantities. This step made use of the existing national LCI lists as well as the approaches in Germany (AgBB) and France (AFSSET). Prior to implementing the decree on labeling of construction and decoration products with regard to their VOC emissions in 2012 in France, AFSSET (now the French Agency for Food, Environmental and Occupational Health & Safety (ANSES)) developed an assessment scheme similar to the German AgBB system which was published in 2009 (AFSSET, 2009). This evaluation scheme contains 165 LCI values for individual VOC. The German and French evaluation schemes shared almost the same list of substances, with some LCI values diverging due to procedural differences in the respective derivations systems. The EU-LCI working group merged the German and French long lists of substances to a EU-LCI master list of 185 compounds to be harmonized.

The next step was the development of a protocol for setting EU-LCI values. This procedure is based on toxicological and risk assessment principles, and consistent with the principles of REACH.¹ The EU-LCI protocol involves three main stages:

- Compilation of the published toxicological data and available reference values from recognized committees on a particular substance.
- Data evaluation and selection of the key study by examining the original literature if necessary and available.
- Derivation of the EU-LCI value on the basis of a total assessment factor in a standardized factsheet.

The detailed protocol for the derivation of EU-LCI values, the first batch of 82 EU-LCI values and their rationales were published in the ECA Report 29 at the beginning of 2013 (ECA Report 29, 2013). The protocol was presented to the European Commission in December 2012 in order to acquire structural and financial support for the completion of the EU-LCI list. However, it was not evident at the time which Commission

¹ REACH: Registration, Evaluation, Authorization and Restriction of Chemicals.

Directorates-General – the Health- or the Building-Sector-would be responsible for the continuation and implementation of the harmonization process. Hence, the Commission did not state its support for this process. Nevertheless, the EU-LCI working group recognized the importance of its task and decided to continue the derivation of EU-LCI values on its own resources. Simultaneously, the working group kept conducting intensive discussions with the European Commission to receive a mandate for its activity.

2.2. First steps in the development of harmonized VOC classes

In 2010 the European Commission started the discussion on the communication of VOC emissions from construction products at expert level. In December 2014 a first draft document entitled “Harmonized EU-VOC Classes: Classification System for VOC Emissions of Construction Products used in the Indoor Environment” was sent to the governments of the EU Member States for discussion. This draft included a basic system for the declaration of VOC classes (Fig. 1) and was proposed by Edmund Vankann within the Expert Group on Dangerous Substances (EGDS, now Subgroup on Dangerous Substances (SGDS)). The SGDS provides technical input for the Commission, in particular for amending mandates to CEN.

A complete harmonized list of EU-LCI values was considered to be essential for the development of VOC classes in this first draft of 2014. This proposal took into consideration the existing national regulations regarding emissions from construction products and the recommendation of the ECA Report 27 that “the LCI approach is currently the most feasible strategy to assess the health effects of compounds from building products” (ECA Report 27, 2012).

In November 2015, the EU-LCI working group finally received the mandate from the European Commission to establish further EU-LCI values and to complete the harmonized list. The completion of the list progressed as documented at https://ec.europa.eu/growth/sectors/construction/eu-lci/values_en. By the end of 2019 EU-LCI values for 152 compounds were established. Around 30 compounds still need to be harmonized.

Since 2015 the German Environment Agency (UBA) has additionally been supporting the completion of the EU-LCI list by financing the compilation of technical dossiers for some of the remaining compounds. These dossiers are prepared by experienced toxicologists and then presented and discussed within the EU-LCI working group. The final decision upon the derivation of an EU-LCI value is taken by the EU-LCI working group. The European Commission also financed 21 technical dossiers between 2016 and 2019.

As a result of the standardized work started in 2005, the EN 16516: 2018-01 Construction products - Assessment of release of dangerous substances - Determination of emissions into indoor air was published in January 2018 (EN 16516, 2018).

Driving force for the harmonization work is the development of labeling requirements for products with respect to indoor air under the CPR. It is likely that any future labeling scheme assessing health effects

will refer to toxicological reference values, such as the published list of EU-LCI values. Generally, the EU-LCI values make sense only in conjunction with a standard on how to technically test product emissions. Here, the EN 16516 standard defines the requirements for testing of the types and amounts of compounds released in environmental chambers and reporting results. However, currently, there is no harmonized procedure for the evaluation of these results. Few EU Member States have specifications or pass/fail systems for the health-related evaluation of VOC emissions from construction products. Additional voluntary labeling schemes with criteria which are not always easy to understand make it very difficult for architects and consumers to choose ‘safe’ (construction) products regarding controlled emissions. Communication of the emission performance for indoor use applying VOC classes on the CE marking of standardized construction products is a good option for implementation of the health requirements stipulated in the CPR. The harmonized EU-LCI list is considered a key element for the establishment of the emission performance. Fig. 2 depicts the need and key benefits of health-based reference concentrations for construction products emissions.

2.3. Proposed delegated legal act on communication of VOC emissions

Since 2014 a system for the declaration of VOC classes has been continuously under discussion mostly at expert level in the SGDS. Various modifications of the first proposal have been undertaken and discussed over the last years. In May 2017 the European Commission presented a draft delegated act on communication of VOC emissions. The delegated act describes how to establish classes of performance in relation to the essential characteristics of construction products under the CPR. It contains three distinct parameters to assess VOC emissions and define classes: EU-LCI, carcinogenic, mutagenic and reprotoxic (CMRs) VOC, and formaldehyde. The document also includes a list with the derived EU-LCI values up to December 2016. The European Commission’s proposal was discussed at expert level in the SGDS working group but also presented at higher political level.

2.4. Criticism on the proposed delegated legal act

The German Environment Agency (UBA) and the AgBB committee rate the European Commission’s draft proposal as insufficient regarding the requirements for protecting the health of building occupants and avoiding hazards from construction product emissions. Important evaluation criteria such as the R-value, limits for TVOC and SVOC emissions, and limits for substances without a EU-LCI values are missing. The minimum requirements for the existing health protection level such as found in Germany cannot be fulfilled with any of the classes foreseen in the European Commission’s proposal. Moreover, the mandated and validated standard test method EN 16516 is ignored in the draft delegated act.

The most important criticism is the missing R-value which accounts for the combined effects of the emitted substances from construction products (sum of all R_i -values). Toxicological information based on scientific knowledge confirm that substance mixtures with differing compositions can have greater toxic effects than the sum of the toxic effects of the individual substances. It has been verified that substances in VOC emissions often have the same mode of action thus increasing the toxic effect in a linear additive mode. One example here is the irritating effects of substances (Nielsen et al., 2007). Adequate protection against such hazards is the use of an additive R-value. The R value as an assessment criterion results from expert judgment and is a transparent and pragmatic standard approach recommended by different acknowledged expert groups (SCHER, SCENIHR, SCCS, 2012; ECA Report 29, 2013) for the evaluation of cumulative health dangers as confirmed in the JRC Science and policy report from 2014 (JRC, 2014). The new WHO instrument for the cumulative assessment of indoor pollutants is also based on an additive approach (WHO, 2019). The mandated and

TVOC main classes		Optional color scheme	Sub-classes for formaldehyde emissions			
increasing emission ↓	A		f1	f2	f3	fn
	B					
	C					
	D					
	E					
			increasing emission →			
NPD : no performance declared						

Fig. 1. Basic system for the declaration of VOC classes (by Edmund Vankann).

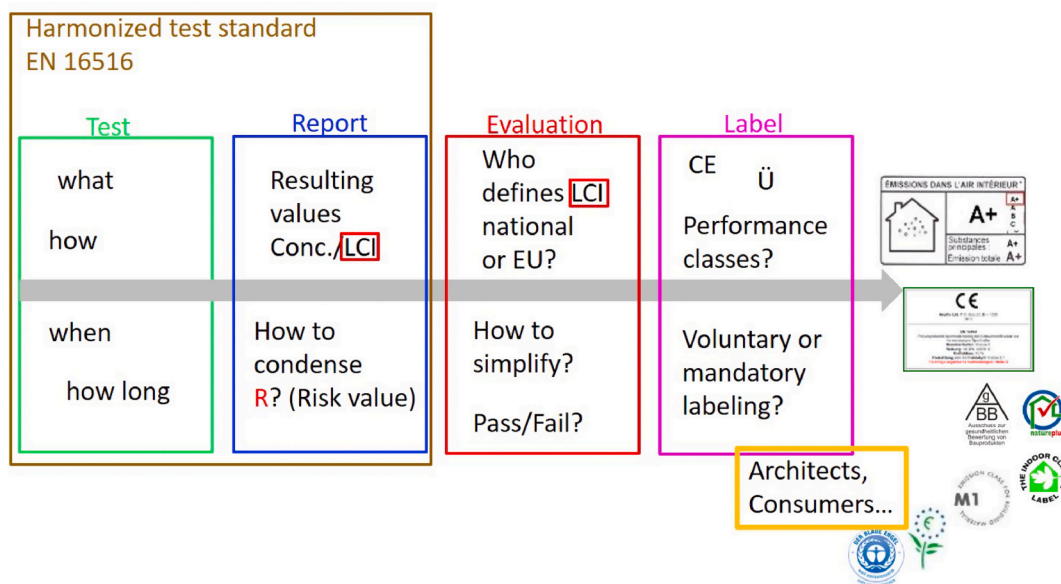


Fig. 2. Data evaluation with EU-LCI values.

validated reference method EN 16516 also includes the R-value criterion as a sum parameter.

By using the proposed VOC classes, the manufacturer cannot declare all relevant performances of a construction product, as the classes do not cover all legally required essential characteristics. This leads to an abolishment of the health protection level for danger defense as defined by the minimum criteria in the AgBB assessment scheme and transposed in the current German legal requirements on health protection for built structures. The proposed criteria make it impossible to select safe products for indoor use. German stakeholders did not accept the EU Commission's proposal and campaigns for resuming negotiations on the technical content of the delegated act. Besides Germany, the EU Member States France, Belgium, and Finland expressed their criticism on the European Commission's proposal as well.

2.5. Alternative proposal by UBA and current discussion

In October 2017 the German Environment Agency (UBA) organized a parliamentary briefing in Brussels to foster the discussion on Europe-wide harmonized labeling of material emissions and facilitate improvements to the draft delegated act (<https://www.umweltbundesamt.de/en/service/dates/briefing-evaluation-of-construction-product>). Experts from national authorities, representatives of the European Commission, Members of the European Parliament, researchers, NGOs and industrial associations participated in this event.

The discussion at the Brussels briefing highlighted the diversity of users of the VOC class concept and the associated communication challenges. Government authorities stated that regulators need a class system that meets national requirements. On the other hand, NGOs expect the VOC classes also to be used by consumers. However, because NGOs are interested above all in ensuring that products are safe, they consider the lack of pass/fail criteria in the European Commission's draft proposal a fatal flaw.

Another issue was the suitability of TVOC as a criterion for the VOC classification system. The Commission's proposal does not include an assessment of TVOC emissions. The omission of this criterion allows, in principle, unlimited emissions from a construction product into indoor air. All speakers acknowledged that TVOC is not a strict toxicological indicator but has served as an effective and necessary parameter being used in building certification and other labeling schemes.

The urgency of adopting a delegated act was emphasized by the participants. Some product standards have been updated and already

comprise the assessment of VOC emissions as an essential characteristic. The first product declarations will soon be available but without a VOC class system, these declarations will contain a long list of substances which layman persons are not expected to understand.

UBA welcomed the opportunity for additional consultations on the draft delegated act and that the information needs of the different stakeholders became clearer. UBA also requested that Member States should be permitted to maintain their national level of protection, especially if this is superior to the system proposed in the delegated act. Table 1 presents UBA's proposal for the improved definition of VOC classes.

The discussions on the draft delegated act are ongoing. An updated version is expected from the EU Commission in 2020.

2.6. Experience from 25 years of material emission labeling in Denmark

The material emission labeling scheme Danish Indoor Climate Labelling was established in 1993 upon a request by the Danish Minister of Housing and Building. The purpose of the labeling scheme is three-fold: to improve the air quality in buildings by documenting the impact of products on the indoor air quality, to contribute to the development of indoor friendly products, and to support the selection of indoor friendly products.

The material emissions are evaluated based on chemical analysis of individual VOC, sensory (odour) evaluation and release of fibres and particles (ceiling systems only). The labeling scheme has proved to be useful in several ways. There is now an increased focus on indoor air and low-emitting materials in Denmark, there has been a tendency towards generally lowered emission levels over the years among the labelled

Table 1

UBA's proposal to define VOC classes using five parameters (Source: Presentation Lilian Busse, Brussels, October 2017).

VOC Classes	Declared value (example)	Emission class	Threshold
R-value	0.4	A	<1.0
TVOC	700 µg/m ³	A	<1000 µg/m ³
∑VOC without EU-LCI	50 µg/m ³	A	<100 µg/m ³
TSVOC	90 µg/m ³	A	<100 µg/m ³
Carcinogens	Not detectable	A	<1 µg/m ³

products, and in many cases the test results have been used actively for product development.

Although the use of the labeling scheme is voluntary, the scheme has been referred to in the Danish Building Code since its establishment. In the Guidance Text of the present version of the Building Code (Danish Transport, Construction and Housing Authority, 2018), the EU-LCI concept is acknowledged as a useful method to evaluate indoor material emissions.

The labeling criteria used by the Danish Indoor Climate Labelling was thoroughly revised early 2018 adopting the EN 16516, ISO 16000-28 (ISO 16000-28, 2012) test methods and evaluation method developed by the German AgBB committee and thereby supporting the harmonization of material emission test and evaluation methods.

3. Conclusions

As of 2019, the EU-LCI working group mandated from the European Commission has derived EU-LCI ('lowest concentration of interest') values for 152 organic compounds typically emitted by construction products. In conjunction with a standardized procedure for test chambers (EN 16516) the EU-LCI values are suitable and handy for the health-related assessment of building product emissions. Around 30 compounds still need to be harmonized.

The EU-LCI values are already being used for assessment of indoor material emissions by German and Belgian national regulations and a number of voluntary labeling schemes in the EU Member States. Although not mandatory the EU-LCI concept is also referred to in the guidance text of the Danish Building Code for evaluation of material emissions.

Harmonization of health requirements for construction products is a common objective for all parties involved. From the German point of view, it is essential that the future VOC classes are compatible with a high level of protection for all European building users. Only safe (construction) products should be placed on the market. The delegated act should consider the existing legal requirements in the various EU Member States. Many industry associations consider the acceptance of the delegated act by the EU Member States as being highly important. Manufacturers prefer an efficient VOC class system rather than declaration of all individual substances emitted by a specific product.

The German authorities welcome and intensively support the activities for Europe-wide harmonized health criteria for the assessment of construction products emissions. However, in order to efficiently avoid unacceptable adverse health effects, the European Commission's VOC class proposal needs to be supplemented.

Declaration of competing interest

The authors have declared no conflict of interest.

Acknowledgement

All members of the EU-LCI group are listed at https://ec.europa.eu/growth/sectors/construction/eu-lci_en.

References

AFSSET, 2009. Procédure de qualification des émissions de composés organiques volatils par le matériaux de construction et produits de décorations. Avis de l'Afsset – Rapport d'expertise collective.

- AgBB (Committee for health-related evaluation of building products), 2018. Health-related evaluation procedure for VOC emissions from building products. https://www.umweltbundesamt.de/sites/default/files/medien/360/dokumente/agbb-bewertungsschema_2018.pdf accessed 27.04.20.
- Beall, J.R., Ulsamer, A.G., 1981. Toxicity of volatile organic compounds present indoors. *Bull. N. Y. Acad. Med.* 57 (10), 978.
- Construction Products Regulation, 2011. Regulation (EU) No 305/2011 of the European Parliament and of the Council of 9 March 2011 Laying Down Harmonized Conditions for the Marketing of Construction Products and Repealing Council Directive 89/106/EEC. OJ L 88/5.
- Danish Transport, Construction and Housing Authority, 2018. Bygningsreglementet. <http://bygningsreglementet.dk> accessed 27.04.20.
- Däumling, C., 2012. Product evaluation for the control of chemical emissions to indoor air – 10 years of experience with the AgBB scheme in Germany. *Clean* 40 (8), 779–789. <https://doi.org/10.1002/clean.201000364>.
- Decree n° 2011-321, 2011. Décret n° 2011-321 du 23 mars 2011 relatif à l'étiquetage des produits de construction ou de revêtements de mur ou de sol et des peintures et vernis sur leurs émissions de polluants volatils. J. Off. de la République Française du 16, 5343, 25 mars 2011.
- Decree [C - 2014/24239], 2014. Royal Decree Establishing Threshold Levels for Emissions in the Indoor Environment of Building Products for Certain Intended Uses. Federal Public Service Health. Food Chain Safety and Environment, Belgium.
- ECA Report 18, 1997. Evaluation of VOC emissions from building products – solid flooring materials. In: Report No. 18, European Commission, Joint Research Centre. Environment Institute. EUR 17334 EN.
- ECA Report 24, 2005. Harmonization of Indoor Material Emissions Labeling Systems in the EU – Inventory of Existing Schemes. Office for Official Publications of the European Communities, Luxembourg, ISBN 92-79-01043-3. EUR 21891 EN.
- ECA Report 27, 2012. Harmonization framework for indoor products labelling schemes in the EU. In: Report No. 27, European Commission, Joint Research Centre. Institute for Health & Consumer Protection. EUR 25276 EN.
- ECA Report 29, 2013. Harmonization framework for health based evaluation of indoor emissions from construction products in the European Union using the EU-LCI concept. In: Report No. 29, European Commission, Joint Research Centre. Institute for Health & Consumer Protection. EUR 26168 EN.
- EN 16516, 2018. Construction Products - Assessment of Release of Dangerous Substances - Determination of Emissions into Indoor Air. EN 16516:2018-01.
- ISO 16000-28, 2012. Indoor Air – Part 28: Determination of Odour Emissions from Building Products Using Test Chambers. ISO 16000-28:2012.
- Joint Research Centre (JRC), 2014. Assessment of Mixtures - Review of Regulatory Requirements and Guidance. JRC Science and Policy Reports. European Commission, Joint Research Centre, Institute for Health and Consumer Protection, Ispra.
- Mølhave, L., 1991. Volatile organic compounds, indoor air quality and health. *Indoor Air* 1, 357–376. <https://doi.org/10.1111/j.1600-0668.1991.00001.x> accessed 27.04.20.
- Müller, B., Mertes, A., 2017. Indoor air quality after installation of building products in energy-efficient buildings. UBA Texte 73/2017, Umweltbundesamt, Berlin. <https://www.umweltbundesamt.de/publikationen/indoor-air-quality-after-installation-of-building>. accessed 27.04.20.
- Nielsen, G.D., Wolkoff, P., Alarie, Y., 2007. Sensory irritation: risk assessment approaches. *Regul. Toxicol. Pharmacol.* 48, 6–18.
- Paciência, I., Madureira, J., Rufo, J., Moreira, A., Fernandes, E.D.O., 2016. A systematic review of evidence and implications of spatial and seasonal variations of volatile organic compounds (VOC) in indoor human environments. *J. Toxicol. Environ. Health, Part B* 19 (2), 47–64.
- Regulation (EC) No 1272/2008, 2008. Regulation (EC) No 1272/2008 of the European Parliament and of the Council of 16 December 2008 on Classification, Labelling and Packaging of Substances and Mixtures, Amending and Repealing Directives 67/548/EEC and 1999/45/EC, and Amending Regulation (EC) No 1907/2006. OJ L 353, 31.12.2008. Last Amended by the Commission Regulation (EU) No 1297/2014. OJ L 350, 6.12.2014.
- SCHER, SCENIHR, SCCS, 2012. Toxicity and Assessment of Chemical Mixtures. Brussels. European Commission. DG Health & Consumers, p. 50 accessed 27.04.20.
- Seppänen, O., 2008. Ventilation strategies for good indoor air quality and energy efficiency. *Int. J. Vent.* 6 (4), 297–306.
- WHO, 2019. Towards a Tool for Assessment of Cumulative Risks from Indoor Air Pollutants in Public Settings for Children: the First Expert Consultation. Meeting Report. WHO Europe, Bonn. http://www.euro.who.int/_data/assets/pdf_file/0020/410780/Indoor-air-pollutants-public-children-first-consultation-report.pdf?ua=1. accessed 27.04.20.
- Wolkoff, P., 1995. Volatile organic compounds sources, measurements, emissions, and the impact on indoor air quality. *Indoor Air* 5 (S3), 5–73.